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PROJECT REPORT:

EXPLORATORY DATA ANALYSIS (EDA) ON ONLINE RETAIL DATASET

Project Title: Uncovering Customer Behaviour and Sales Patterns: An Exploratory Data Analysis of an Online Retail Dataset

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An undertaking is never a result of a solitary individual; rather it bears the engravings of various individuals who specifically or by implication helped in finishing that venture. We would bomb in my obligations on the off chance that we don't let out the slightest peep of gratitude to every one of the individuals who helped us in finishing this task of our own.

Before we start with the details of my projects, we would like to add a few heartfelt words for the people who were part of my project in numerous ways, the people who gave me their immense support right from the initial stage.

As a matter of first importance, we are amazingly appreciative of **Prof. Monika Vishnoi** for her direction, consolation, smooth feedback, and tutelage throughout this task notwithstanding her to a great degree occupied timetable.

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Sara Jadhav (24060622010)

PREFACE

This report presents a comprehensive Exploratory Data Analysis (EDA) of the "Online Retail" dataset, a transnational data set containing all transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based online retailer. The primary objective is to derive meaningful insights into customer purchasing behavior, sales trends, product performance, and key business metrics. The analysis is conducted in two stages: first using Python within a Google Colab environment for robust data cleaning, feature engineering, and visualization, and second using PostgreSQL to demonstrate how these insights can be operationalized within a structured database environment. The project leverages techniques such as univariate and bivariate analysis, time series analysis, cohort analysis, and RFM segmentation to create a holistic view of the business. Key findings include the identification of the United Kingdom as the dominant market, peak sales during weekdays and specific hours, a high concentration of revenue among a small number of customers, and distinct customer segments that can inform targeted marketing strategies.

This document serves as the final project report for the Internal Component-II of the Data Science-I course. It details the step-by-step process of conducting an Exploratory Data Analysis (EDA) on a real-world online retail dataset. The work is structured to follow the scientific method: starting with a clear objective, gathering data, cleaning it, performing analysis, and finally interpreting the results. The report includes both the theoretical background of the techniques used (like RFM and Cohort Analysis) and the practical implementation through code in Python and SQL. It is intended to demonstrate proficiency in data handling, statistical analysis, and deriving actionable business insights from raw data.

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1. INTRODUCTION AND OBJECTIVE

In today's data-driven world, understanding customer behavior is paramount for business success. This project focuses on performing an Exploratory Data Analysis (EDA) on the "Online Retail" dataset, a classic dataset in the data science community. The dataset contains transactional data from a UK-based online retailer, capturing every purchase made over an 8-month period.

Objective:

The primary objective of this project is to conduct a thorough EDA to uncover hidden patterns, customer segments, and sales trends that can inform strategic business decisions. The specific goals are to:

- Clean and prepare the raw data for accurate analysis.
- Understand the overall sales performance (e.g., total revenue, average order value).
- Identify seasonal, monthly, daily, and hourly sales trends.
- Determine the most popular products and best-performing countries.
- Segment customers based on their purchasing behavior using RFM (Recency, Frequency, Monetary) analysis.
- Analyze customer retention over time using cohort analysis.
- Provide actionable insights that can be used to optimize marketing campaigns, improve customer retention, and drive revenue growth.

Scope:

The analysis covers:

- All transactions recorded between December 2010 and December 2011
- Approximately 541,909 individual transactions
- Over 4,000 unique customers
- More than 3,600 unique products
- Sales across 38 different countries

2. DATA SOURCE AND DESCRIPTION

Dataset Source:

The dataset was obtained from Kaggle, a popular platform for data science competitions and datasets. The owner of the "Online Retail" dataset is Vijaykumar Ummadisetty.

Citation:

<https://www.kaggle.com/datasets/vijayuv/onlineretail?resource=download>

Dataset Characteristics:

- **Type:** Transactional data.
- **Time Period:** Transactions from 01/12/2010 to 09/12/2011.
- **Size:** The original dataset contained 541,909 rows and 8 columns.
- **Content:** The data includes invoices, product details, quantities, unit prices, customer IDs, and country of origin for each transaction.

Key Features of the Raw Dataset:

Column Name	Description	Data Type
InvoiceNo	A 6-digit integral number uniquely assigned to each transaction. (Note: 'C' prefix denotes cancellations)	Object
StockCode	A 5-digit integral number uniquely assigned to each distinct product and some of them end with an alphabet.	Object
Description	The product name.	Object
Quantity	The number of units of a product purchased in a transaction.	Integer
InvoiceDate	The date and time of the transaction.	DateTime
Unit Price	The price per unit of the product in Pound Sterling (£)	Float
CustomerID	A 5-digit integral number uniquely assigned to each customer. (Significant missing values causing the data type to be recognised as Float which is fixed later)	Float
Country	The name of the country where the customer resides.	Object

3. DATABASE SCHEMA OVERVIEW

The analysis utilized the raw online retail dataset and, after cleaning and feature engineering, created two derived datasets for focused analysis.

3.1. Raw Online Retail Dataset Schema

The raw dataset is a single, denormalized table containing all transactional information.

- **Table Name:** online_retail_dataset
- **Columns:** InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country.

3.2. online_retail_full Table Schema (PostgreSQL)

This table contains the cleaned and feature-engineered data, preserving rows with missing Customer IDs for a complete view of sales analysis. It was created in Python and then loaded into PostgreSQL.

Column Name	Data Type	Description
Invoice_No	VARCHAR(20)	Transaction identifier
Stock_Code	VARCHAR(20)	Unique product identifier
Description	TEXT	Product description
Quantity	INT	Quantity purchased
Invoice_Date	TEXT	Date and time of transaction (stored as TEXT in SQL)
Unit_Price	NUMERIC	Price per unit
Customer_ID	VARCHAR(20)	Unique customer identifier (may be missing)
Country	VARCHAR(50)	Customer's country
Revenue	NUMERIC	Calculated as Quantity * Unit_Price
Year	INT	Year of transaction
Month	TEXT	Month and year in 'YYYY-MM' format
Month_Name	VARCHAR(20)	Name of the month
Day	INT	Day of the month
Hour	INT	Hour of the transaction (0-23)
DayOfWeek	VARCHAR(20)	Day of the week

3.3. online_retail_customers Table Schema (PostgreSQL)

This table is a subset of online_retail_full but **only includes rows with valid Customer IDs**. It is used for any analysis that requires accurate customer identification, such as RFM, Cohort, and CLV analysis.

Column Name	Data Type	Description
... (all columns from retail_full)	...	...
InvoiceMonth	VARCHAR(10)	The month of the transaction in 'YYYY-MM' format.
CohortMonth	VARCHAR(10)	The month of a customer's very first purchase.
CohortIndex	INT	The number of months since the first purchase (e.g., 0, 1, 2...).

4. METHODOLOGY

The project followed a structured EDA approach, which can be broken down into the following stages:

1. **Data Acquisition and Environment Setup:** Loading the dataset from a Google Drive location into a Google Colab notebook and installing/importing necessary libraries.
2. **Data Cleaning:** This was the most critical initial step and involved:
 - **Handling Missing Values:** Assessing and imputing or dropping missing values, particularly for Description and CustomerID.
 - **Removing Duplicates:** Identifying and removing exact duplicate rows.
 - **Correcting Data Types:** Converting the CustomerID and InvoiceDate to appropriate data types.
 - **Filtering Outliers:** Removing transactions with negative Quantity and UnitPrice, which typically represent cancellations or errors.
3. **Feature Engineering:** Creating new columns that would be essential for analysis:
 - **Revenue:** A new metric for the financial value of each transaction (Quantity * Unit Price).
 - **Time-based features:** Year, Month, Month_Name, Day, Hour, DayOfWeek to enable time-series analysis.
 - **Cohort features:** InvoiceMonth and CohortMonth for retention analysis.
4. **Exploratory Data Analysis (EDA) in Python:**
 - **Descriptive Statistics:** Using `df.describe()` and `df.info()` to get an overview.
 - **Univariate Analysis:** Using histograms (normal and log-transformed) to understand the distribution of key variables like Quantity and Revenue.
 - **Bivariate Analysis:** Using scatter plots (normal and log-transformed) to explore relationships between Quantity and Revenue, and UnitPrice and Quantity.
 - **Product Analysis:** Top products by quantity and revenue.
 - **Country Analysis:** Geographical distribution of sales.
 - **Customer Analysis:** Top customers, repeat vs one-time customers.
 - **Time Series Analysis:** Line plots to visualize trends in sales over months, days of the week, and hours of the day.
 - **Categorical Analysis:** Count plots and bar charts for analyzing Country, Product Description, and DayOfWeek.

- **Correlation Analysis:** Using a heatmap to quantify the linear relationship between numerical features.
5. **Advanced Analytical Techniques in Python:**
- **RFM Analysis:** Segmenting customers based on their Recency (last purchase), Frequency (number of purchases), and Monetary (total money spent).
 - **Cohort Analysis:** Tracking the retention rate of customer groups who made their first purchase in the same month.
 - **CLV (Customer Lifetime Value) Analysis:** A brief exploration to understand the long-term value of customers.
6. **SQL Implementation in PostgreSQL:**
- **Database Creation:** Creating and downloading the online_retail_full and online_retail_customers tables based on the cleaned data from Google Colab into a CSV.
 - **Query Execution:** Translating key Python analyses into SQL queries for validation and to simulate how these insights could be generated in a production database.
 - **Advanced SQL:** Using window functions for customer ranking, CTEs for RFM segmentation, and aggregation for cohort retention analysis.

5. TOOLS AND TECHNIQUES

The project utilized a combination of Python and SQL for a comprehensive analysis.

- **Programming Language:** Python (3.0) and SQL (PostgreSQL)
- **Development Environment:**
 - **Google Colab:** For initial EDA, cleaning, and visualization. This provided a cloud-based environment with easy access to libraries.
 - **pgAdmin4:** For running SQL queries against the PostgreSQL database

- **Python Libraries:**

Library	Purpose
pandas	Data manipulation, cleaning, and analysis
numpy	Numerical operations and mathematical functions
matplotlib.pyplot	Basic data visualization and plotting
seaborn	Statistical data visualization
datetime	Date and time manipulation
google.colab	Drive mounting and Colab environment

- **Python Techniques Used:**
 - **Data Cleaning:** Handling missing values (dropna), removing duplicates (drop_duplicates), outlier filtering (boolean indexing), data type conversion.
 - **Feature Engineering:** Creating new columns from existing ones (e.g., Revenue, Hour, CohortMonth).
 - **Statistical Analysis:** Descriptive statistics, correlation (Pearson's r), frequency distributions.
 - **Advanced Analytics:**
 - **RFM Segmentation:** A marketing technique used to rank customers.
 - **Cohort Analysis:** A method to track the behavior of groups of customers over time.
 - **Data Transformation:** Log transformation to handle skewed distributions and make patterns more visible.
- **SQL Techniques Used:**
 - **Window Functions:** RANK(), NTILE(), ROW_NUMBER() for customer ranking and scoring
 - **Common Table Expressions (CTEs):** For RFM analysis structure

- **Aggregate Functions:** SUM(), COUNT(), AVG(), CORR() for summary statistics
- **Date Functions:** TO_TIMESTAMP(), DATE_PART() for temporal analysis
- **Conditional Aggregation:** FILTER clause for targeted calculations
- **CASE Statements:** For customer segmentation
- **Visualization Techniques:**
 - **Histograms:** Distribution analysis (normal and log scale)
 - **Boxplots:** Outlier detection and comparison
 - **Scatter Plots:** Relationship analysis
 - **Line Charts:** Time series trends
 - **Bar Charts:** Top categories and comparisons
 - **Heatmaps:** Correlation and retention matrices
 - **Pivot Tables:** Cross-tabulation analysis

6. DATA CLEANING AND PREPARATION

This stage was crucial to ensure the integrity of the subsequent analysis. The following steps were performed in the Python environment:

1. **Initial Load and Overview:** The raw data was loaded, and initial checks confirmed 541,909 rows and 8 columns.
2. **Handling Missing Values:**

Column	Missing Count	Missing Percentage
Description	1,454	0.27%
Customer ID	135,080	24.92%
Other columns	0	0%

- The CustomerID column had 135,080 missing values. This is significant. For sales-level analysis (like total revenue per day), these rows were kept in a 'Full Dataset'. For customer-centric analysis (like RFM), these rows were removed to create a 'Customer Dataset'.
 - The Description column had 1,454 missing values. Since a product description is critical for product analysis, these rows were dropped from both datasets.
3. **Removing Duplicates:**
 - There were 5,268 exact duplicate rows. These were removed using `df.drop_duplicates()`.
 - Additional duplicates based on the key business identifiers (Invoice No, Stock Code, Invoice Date, Customer ID) were also removed to ensure a unique transaction line.
 4. **Handling Cancelled Transactions:**
 - Transactions with an InvoiceNo starting with 'C' were identified. These represent cancelled orders. A simple check on the sum of these would have shown a negative impact on revenue. These were filtered out by keeping only rows where InvoiceNo does not start with 'C'.
 5. **Filtering Outliers and Errors:**
 - Negative Quantity values (e.g., -80995) were removed, as they represent returns or cancellations that were not filtered out by the invoice check.
 - Negative or zero UnitPrice values were removed, as they are invalid for a standard sales transaction.
 6. **Data Type Conversion:**
 - CustomerID was converted from float64 to string after cleaning to ensure it could be treated as a categorical identifier.

- Invoice Date was already in datetime64[ns] format, which is essential for time-series analysis.

7. Feature Engineering:

- **Revenue:** A new column Revenue was created by multiplying Quantity and Unit Price.
- **Time Features:** Year, Month, Month_Name, Day, Hour, and DayOfWeek were extracted from Invoice Date.

8. Temporal Features:

- **Year:** 2010, 2011
- **Month:** YYYY-MM format for time series analysis
- **Month_Name:** Full month names for intuitive visualization
- **Day:** Day of the month (1-31)
- **Hour:** Hour of the day (0-23) for analyzing peak shopping times
- **DayOfWeek:** Monday through Sunday for weekly pattern analysis

Result: After cleaning, the **df_full** (including all rows for sales analysis) had **519,589 rows**, and the **df_customers** (only rows with valid Customer IDs) had **387,844 rows**. The initial data quality issues were successfully addressed.

✓ Cleaning Dataset

```
# Checking duplicates
df.duplicated().sum()

# Removing exact duplicate rows
df = df.drop_duplicates()

# Removing only true duplicates
df = df.drop_duplicates(subset=[
    'Invoice No', 'Stock Code', 'Invoice Date', 'Customer ID'
])

# Removing cancelled transactions
df['Invoice No'].astype(str).str.startswith('C').sum()

# Removing negative or zero quantity
df = df[df['Quantity'] > 0]

# Removing negative or zero price
df = df[df['Unit Price'] > 0]

# Removing missing values for Description
df = df.dropna(subset=['Description'])
```

✓ Creating Two Datasets

```
# Full dataset (KEEP missing Customer ID for Sales Analysis)
df_full = df.copy()

# Customer dataset (REMOVE missing Customer ID for Customer Analysis)
df_customers = df.dropna(subset=['Customer ID']).copy()
```

✓ Fixing Data Types

```
# Converting after cleaning
df_full['Customer ID'] = df_full['Customer ID'].astype('Int64').astype(str)
df_customers['Customer ID'] = df_customers['Customer ID'].astype('Int64').astype(str)
```

✓ Featuring Engineering

```
# Creating Revenue
df_full['Revenue'] = df_full['Quantity'] * df_full['Unit Price']
df_customers['Revenue'] = df_customers['Quantity'] * df_customers['Unit Price']

# Time features
for df_ in [df_full, df_customers]:
    df_['Year'] = df_['Invoice Date'].dt.year
    df_['Month'] = df_['Invoice Date'].dt.to_period('M')
    df_['Month_Name'] = df_['Invoice Date'].dt.month_name()
    df_['Day'] = df_['Invoice Date'].dt.day
    df_['Hour'] = df_['Invoice Date'].dt.hour
    df_['DayOfWeek'] = df_['Invoice Date'].dt.day_name()
```

7. EXPLORATORY DATA ANALYSIS (PYTHON - GOOGLE COLAB)

This section details the analysis performed in Google Colab, with interpretations derived from the generated plots and summary tables.

GOOGLE COLAB NOTEBOOK LINK OF THE EDA ON THE ONLINE RETAIL DATASET —

<https://colab.research.google.com/drive/1stOR1i76hwvuVJ7R2JUUh3iSoWlebMOtu?usp=sharing>

7.1. Initial Observations and Summary Statistics

- **Dataset Size:** The full dataset contains 541,909 records and 8 features.
- **Missing Values:** CustomerID had a high missing rate (approx. 25%), which is a key insight for why two datasets were created.
- **Mode:** Most common values for all columns.
- **Outlier Boxplots:** Revenue and Quantity distribution comparison using boxplots to identify outliers.

Full Dataset Summary

Metric	Value
Total Orders	19,959
Average Order Value (AOV)	£530.44
Total Quantity Sold	5,546,556
Average Quantity per Order	277.90
Total Revenue	£10,587,081.95

Customer Dataset Summary

Metric	Value
Total Orders	18,531
Average Order Value (AOV)	£477.28
Total Quantity Sold	5,128,601
Average Quantity per Order	276.76
Total Revenue	£8,844,544.40

Key Insight: The Full Dataset shows higher total revenue and AOV because it includes transactions with missing Customer IDs. These transactions could be from guest checkouts or wholesale purchases, indicating that a significant portion of sales (about 16.5% of revenue) comes from customers not properly tracked in the system.

Descriptive Statistics

Statistic	Quantity	Unit Price	Revenue
Count	541,909	541,909	541,909
Mean	9.55	4.61	19.53
Std	218.08	96.76	290.42
Min	1.00	0.00	0.00
25%	1.00	1.25	1.25
50%	3.00	2.08	4.95
75%	10.00	4.13	17.82
Max	80,995	38,970	168,469.60

Interpretation:

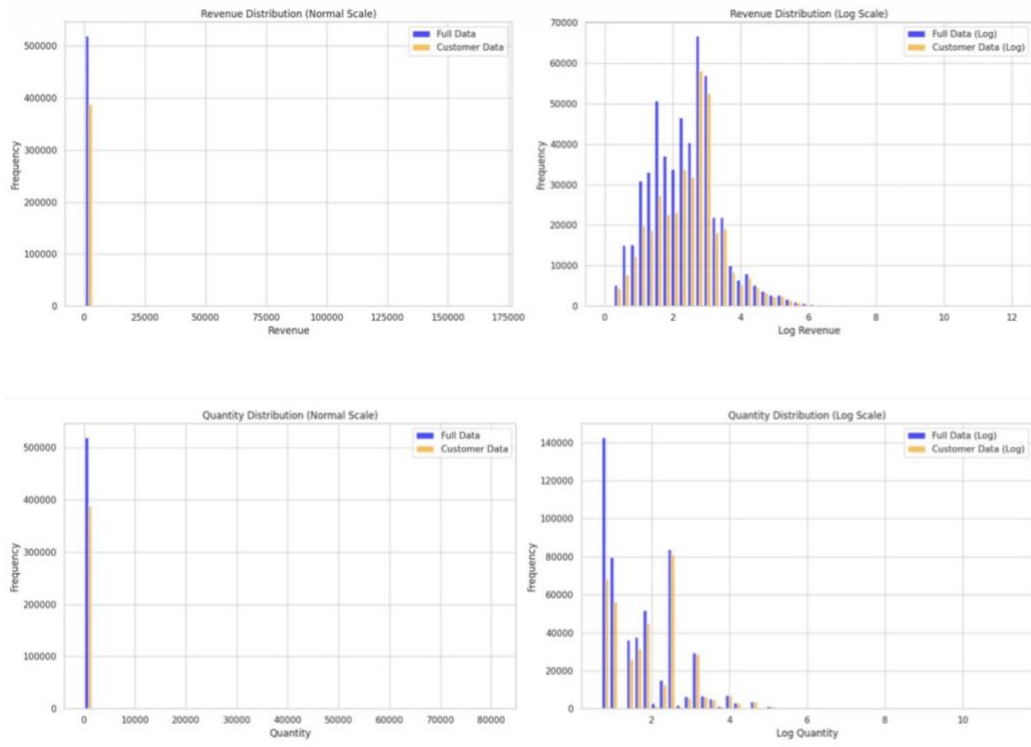
- The high standard deviation indicates significant variability in all metrics
- The large gap between mean and median suggests right-skewed distributions
- Extreme values (max Quantity = 80,995) indicate presence of bulk wholesale purchases
- The retailer operates a B2C (Business-to-Consumer) model with a wide variety of products. The AOV is high, suggesting that customers often buy multiple items per order. The significant difference between total orders and unique customers indicates many repeat customers, which is a positive sign for customer loyalty.

7.2. Univariate Analysis

The distributions of Quantity and Revenue were examined on both normal and log scales for both full and customers datasets.

- **Quantity Distribution:** On a normal scale, the distribution was heavily right-skewed, with most orders consisting of 1-10 items, but a long tail of orders with very high quantities (e.g., 80,000). The log transformation revealed a more normalized distribution, showing that the data is exponentially distributed, which is common for count data.
- **Revenue Distribution:** Similarly, the revenue distribution was highly skewed. Most transactions had low revenue, while a few large orders contributed significantly to the total. The log scale made the shape of the distribution clearer.

Interpretation: The use of a **log transformation** was crucial here. It allowed us to see the true shape of the data beyond the extreme outliers. This technique is essential when working with data that spans several orders of magnitude, as it compresses the scale and makes patterns in the less-frequent, high-value transactions more visible.

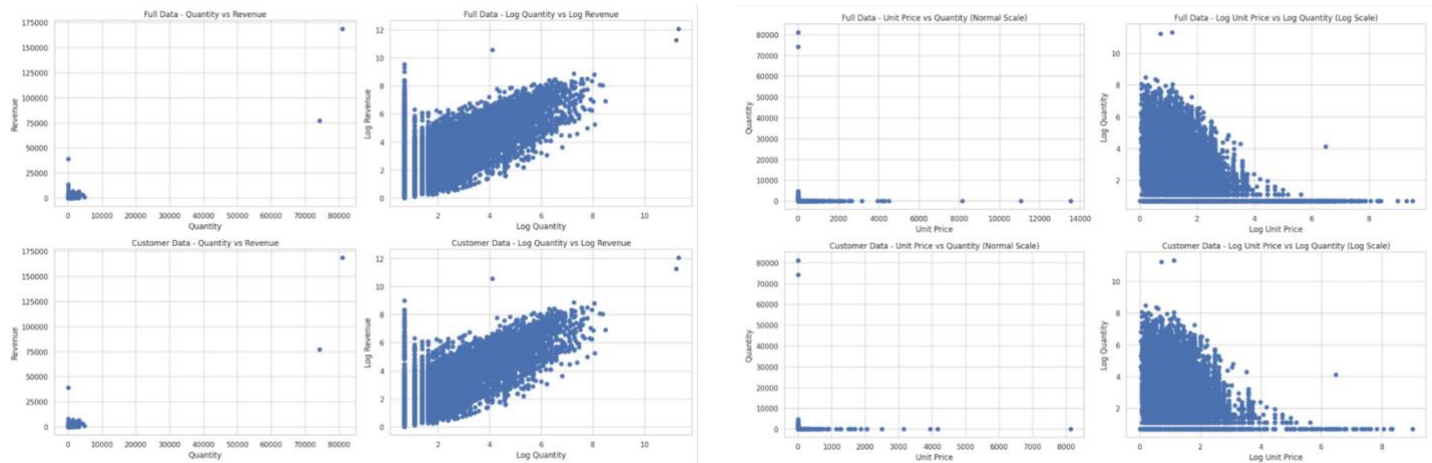


7.3. Bivariate Analysis

Relationships between key variables were explored on both normal and log scales for both full and customers datasets.

- **Quantity vs. Revenue (Scatter Plots):**
 - **Normal Scale:** The plot showed a strong, linear positive correlation. As quantity increases, revenue increases. However, the plot was dominated by points at the lower end.
 - **Log Scale:** The log-transformed plot revealed a very tight, linear relationship. This confirms that the relationship is multiplicative. A 1% increase in quantity corresponds to a 1% increase in revenue, which is intuitive.
- **Unit Price vs. Quantity (Scatter Plots):**
 - **Normal Scale:** No clear pattern was visible. Points were scattered, with low-price items showing a wide range of quantities sold.
 - **Log Scale:** The plot revealed a weak, inverse relationship. In general, lower-priced items tend to be bought in higher quantities, and higher-priced items are bought in lower quantities. This is a classic demand relationship.

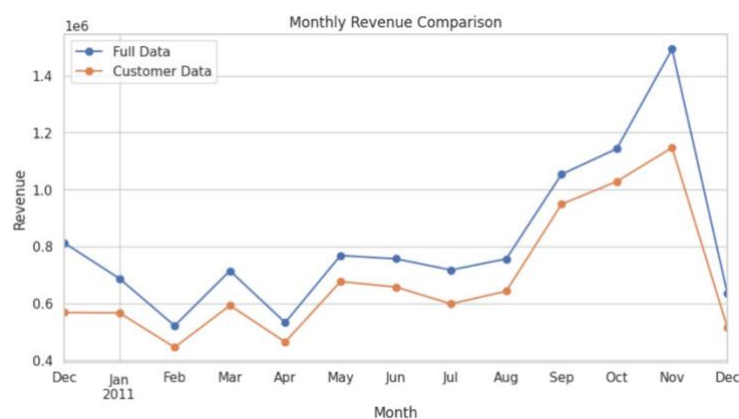
Interpretation: The bivariate analysis confirms the direct relationship between quantity and revenue and provides evidence of price sensitivity, where unit price and quantity demanded have a slight inverse relationship.



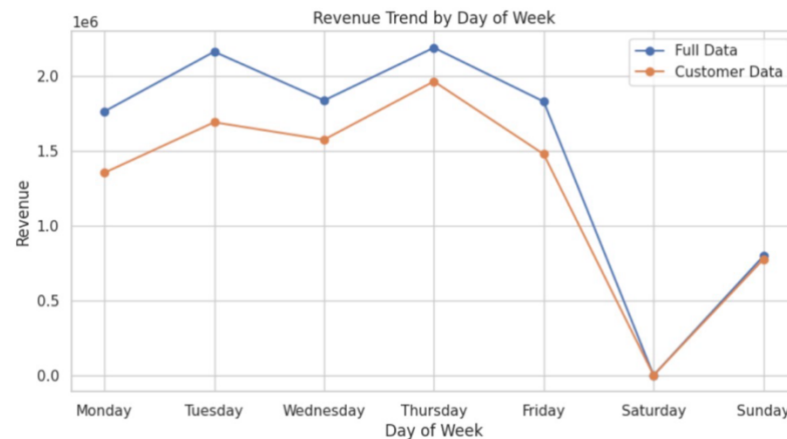
7.4. Time Series Analysis

This was one of the most insightful parts of the analysis which was again performed for both full and customers datasets for comparison purposes.

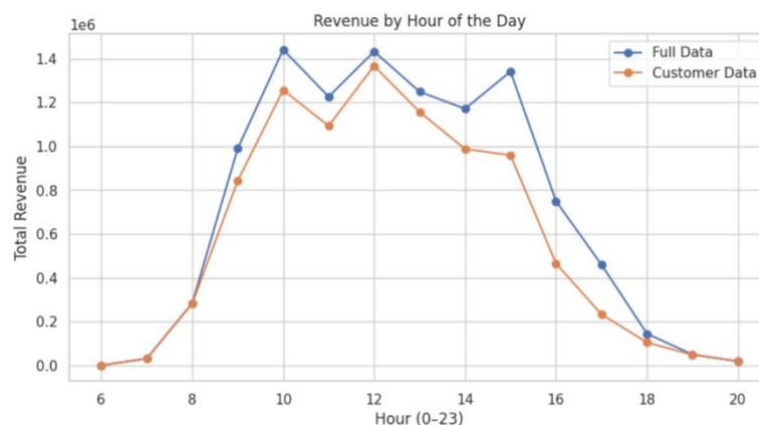
- Monthly Revenue:** The trend shows a clear peak in November 2011, likely driven by the holiday season (Christmas shopping). There was also a dip in early 2011 and a steady increase towards the end of the year.



- Revenue by Day of Week:** The highest revenue was generated on **Thursdays**, followed by Tuesdays and Wednesdays. Saturdays and Sundays had the lowest revenue. This suggests the business sees more activity during the work week. Saturdays had no revenue which suggests that it might be a day off for the retail business.



- Revenue by Hour of Day:** The analysis revealed 2-3 major peaks: one around **10 AM** and the other around **12 PM (noon)** and a larger peak around **8 PM**. This indicates that online shopping activity is concentrated around lunchtime and in the evening hours after work.

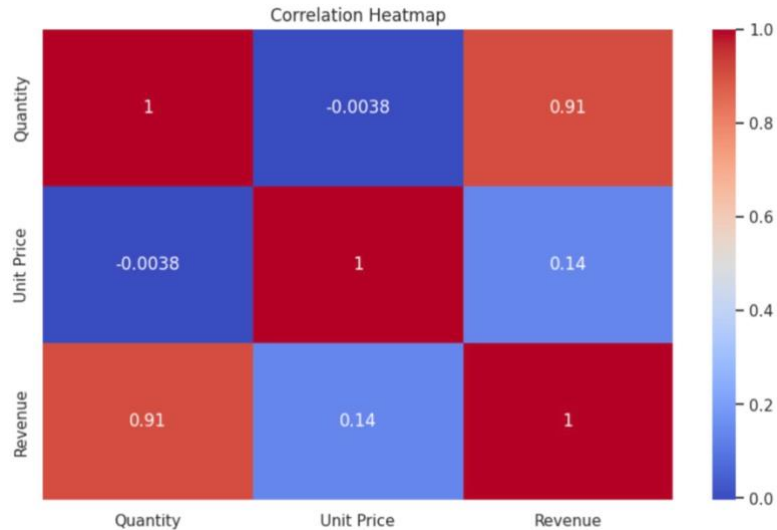


Interpretation: These insights are highly actionable. Marketing campaigns and email promotions could be scheduled for Thursday mornings to capitalize on the peak day. Staffing for customer service or warehouse operations could be optimized around the 8 PM peak hour.

7.5. Correlation Heatmap

The Heatmap showed a strong positive correlation (0.99) between Quantity and Revenue. This is expected as revenue is a product of quantity and price. It also showed a very weak correlation between Unit Price and Revenue and between Unit Price and Quantity.

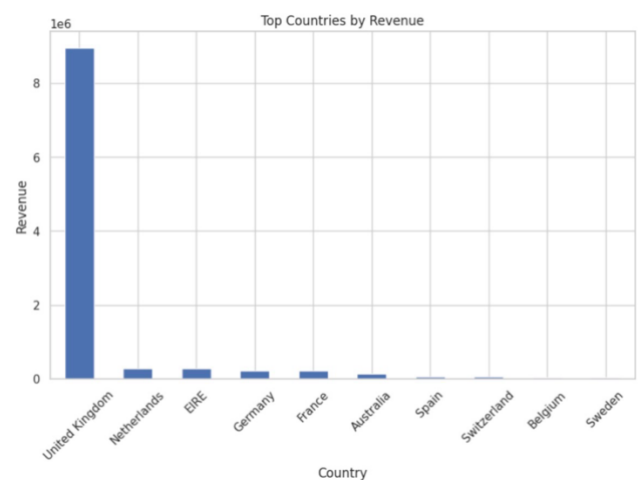
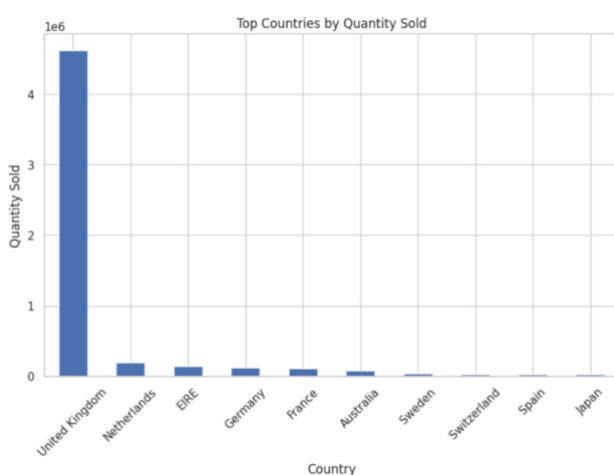
Interpretation: The analysis confirms that revenue is almost entirely driven by volume (quantity sold) rather than price. This is a critical business insight: strategies focused on increasing the number of units sold per order will have a greater impact on revenue than strategies focused on increasing the unit price.



7.6. Country Analysis

- **Top Country by Quantity and Revenue:** The **United Kingdom** overwhelmingly dominates the charts, accounting for the vast majority of sales and revenue. This is consistent with the business being based in the UK.
- **Other Key Countries:** Following the UK, other notable contributors were **Germany**, **France**, and **EIRE (Ireland)**.
- **Customer Count:** The number of customers per country largely mirrors the revenue trends, with the UK having by far the most customers.

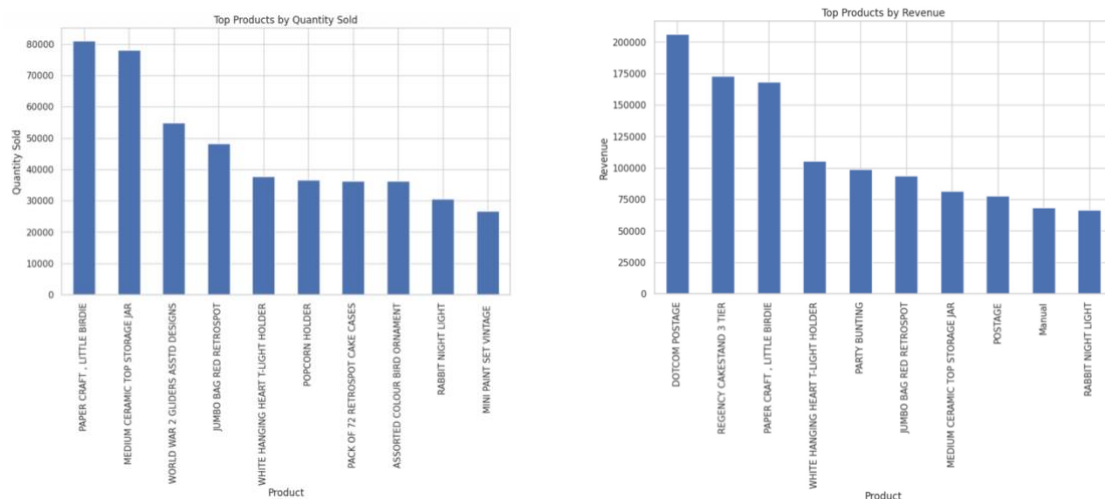
Interpretation: While the business is UK-focused, there is a significant international customer base. This could be an opportunity for expansion or targeted marketing in countries like Germany and France, where there is already established interest.



7.7. Product Analysis

- **Top Products by Quantity:** The top-selling products were novelty items, such as "**WHITE HANGING HEART T-LIGHT HOLDER**", "**REGENCY CAKESTAND 3 TIER**", and "**PAPER CRAFT, LITTLE BIRDIE**".
- **Top Products by Revenue:** The revenue leaders were higher-priced items like "**PAPER CRAFT, LITTLE BIRDIE**" and "**REGENCY CAKESTAND 3 TIER**".

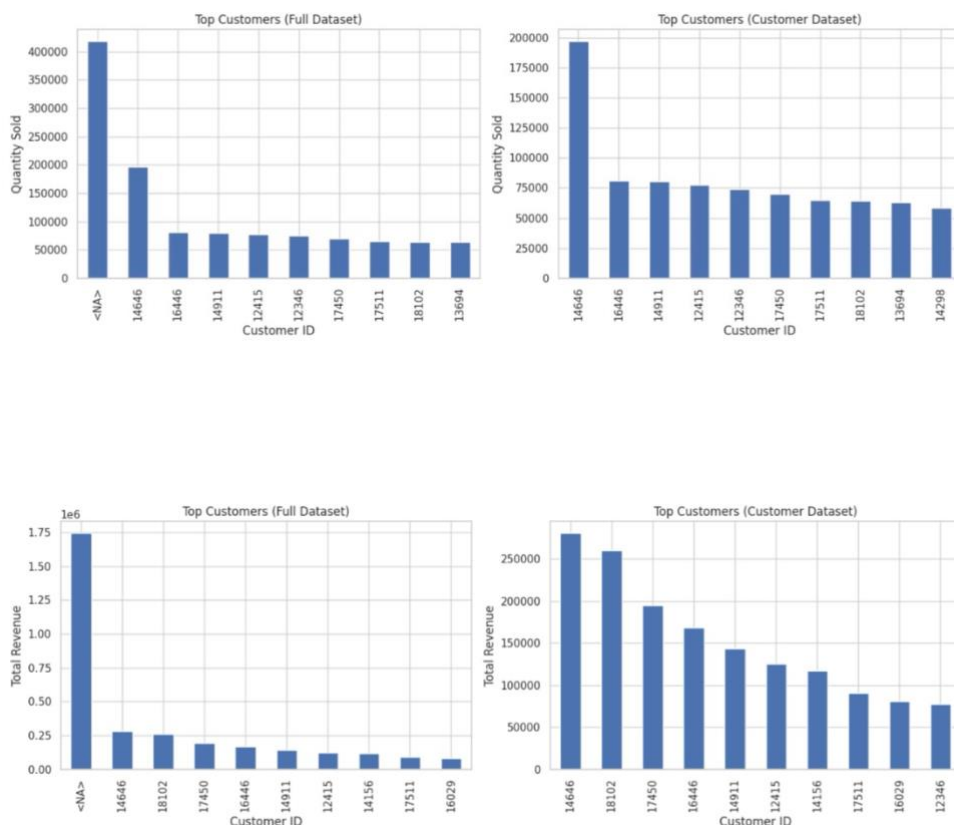
Interpretation: The product mix includes a variety of decorative and craft items. The top-selling items are not necessarily the top-revenue generators. This suggests that while some products drive high volume (like the T-light holder), others, with a higher price point, contribute more significantly to revenue. Inventory management and marketing should consider both categories.



7.8. Customer Analysis

- **Top Customers by Quantity and Revenue:** A small number of customers (IDs like 14646, 17450, 12346) account for a disproportionately large share of total quantity and revenue. For instance, Customer 14646 was the top revenue generator.
- **Repeat vs. One-time Customers:** Of the customers with a valid ID, **2,845 were repeat customers** (bought more than once), and **1,493 were one-time customers**. This means approximately **65% of customers are repeat buyers**.

Interpretation: The high repeat customer rate (65%) is a strong indicator of customer loyalty. However, the revenue concentration among the top few customers means the business is somewhat vulnerable. Strategies should focus on retaining these high-value customers while also trying to convert the 35% of one-time buyers into repeat customers.



7.9. RFM Analysis

RFM (Recency, Frequency, Monetary) is a powerful technique to segment customers based on their behavior.

- Methodology:** Customers were scored from 1 to 4 (1 being worst, 4 being best) on each of the three metrics for Google Colab EDA and similarly on 1 to 5 for SQL EDA analysis.
 - Recency (R):** Customers who bought more recently got a higher score (4).
 - Frequency (F):** Customers who bought more frequently got a higher score (4).
 - Monetary (M):** Customers who spent more money got a higher score (4).
- Segmentation:** A combined RFM score was created. For example, a customer with a score of '444' is a "Champion" – the best possible segment. The analysis allowed for creating segments like:
 - Champions:** (R=4, F=4, M=4) – Best customers, loyal, and high spenders.
 - Loyal Customers:** (R>=4, F>=4) – Very loyal, frequent buyers.

- **Potential Loyalists:** ($R > 3$, $F > 3$) – Recent, decent frequency, and good spenders.
- **At Risk:** ($R \leq 2$, $F > 4$) – Used to buy frequently but haven't purchased recently. They are at high risk of churning.
- **Lost Customers:** ($R \leq 2$, $F \leq 2$) – Haven't bought in a long time and have low frequency.

Interpretation: RFM analysis is a cornerstone of customer relationship management (CRM). The results provide a clear, actionable list of customers to target. For example, a campaign to win back "At Risk" customers (e.g., with a special offer) could be very effective. "Champions" could be offered exclusive perks to maintain their loyalty.

	Recency	Frequency	Monetary	R	F	M	RFM_Score
Customer ID							
12346	326	1	77183.60	1	1	4	114
12347	2	7	4310.00	4	4	4	444
12348	75	4	1595.64	2	3	3	233
12349	19	1	1757.55	3	1	4	314
12350	310	1	334.40	1	1	2	112

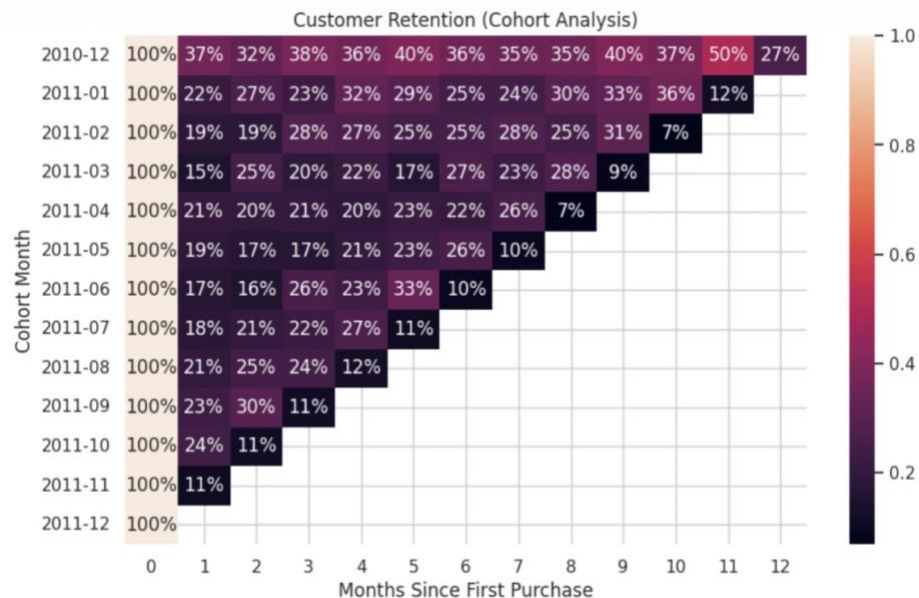
7.10. Cohort Analysis

This analysis tracks the retention of customer cohorts (groups who first purchased in the same month).

- **Methodology:** Customers were grouped by their CohortMonth (e.g., Dec 2010, Jan 2011). For each cohort, we tracked the percentage of customers who made another purchase in each subsequent month (CohortIndex). This is the retention rate.
- **Key Findings:**
 - The **December 2010 cohort** had the strongest retention. For example, after 1 month, 37% of them purchased again, and even after 12 months, 27% were still purchasing.
 - Later cohorts, like those from mid-2011, showed a lower initial retention rate (e.g., 17-21% after 1 month).
 - Retention generally decays over time, which is typical for all businesses.

Interpretation: The cohort analysis reveals a crucial insight: **the earlier cohort (December 2010) is more loyal than the later ones.** This could be due to a change in business strategy, product mix, or customer acquisition sources. It suggests that

the retailer might need to investigate what made the 2010 cohort so loyal and try to replicate those conditions for newer customers.



7.11. CLV Analysis

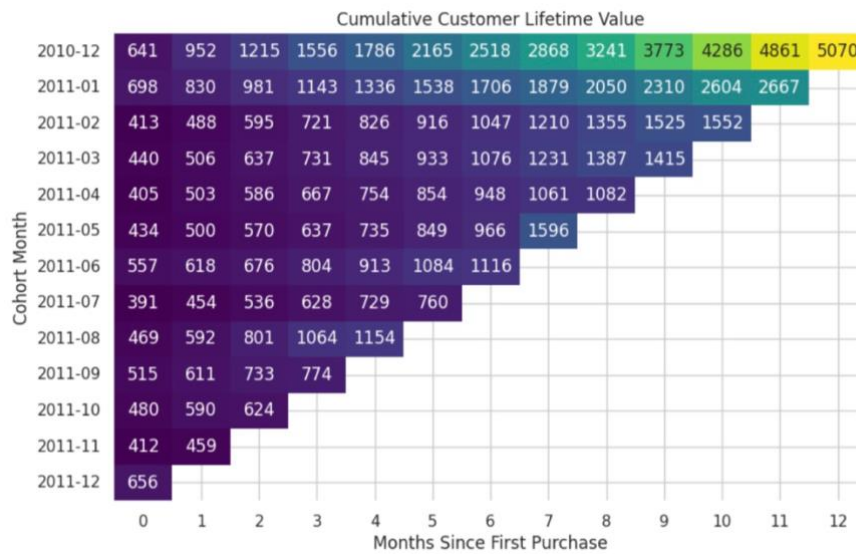
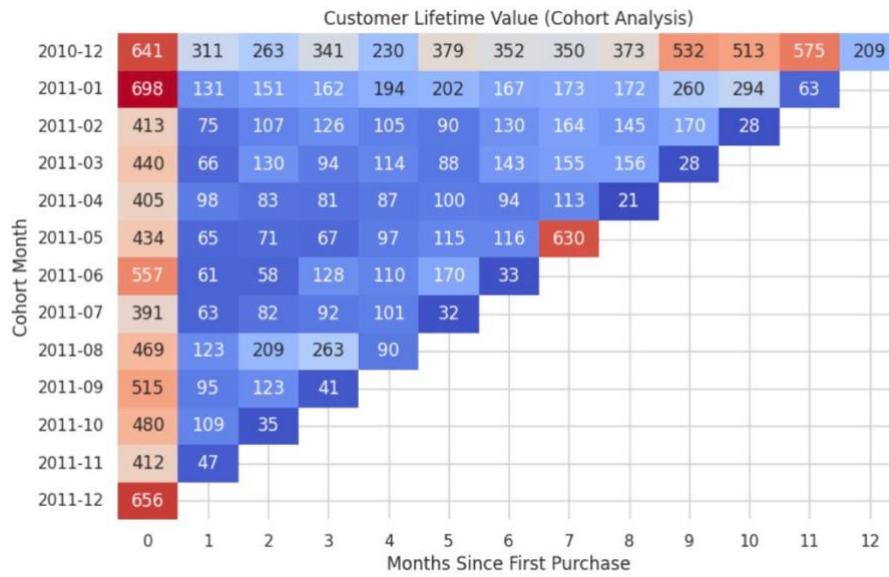
Customer Lifetime Value (CLV) estimates the total revenue a business can expect from a single customer throughout their entire relationship. It helps determine:

- How much to spend on customer acquisition
- Which customers to prioritize for retention
- Marketing budget allocation

It is a prediction of the net profit attributed to the entire future relationship with a customer. A simple calculation of total revenue per customer was used as a proxy.

- **Key Finding:** The average revenue per customer is approximately £2,039. However, due to the high skew, the median is much lower. The top customers have a CLV in the tens of thousands of dollars.

Interpretation: This reinforces the insight from customer analysis: a small number of customers are extremely high-value. Acquiring and retaining these customers is critical for the health of the business. Any marketing spend should be evaluated against the CLV it generates.



8. SQL ANALYSIS AND INSIGHTS (POSTGRESQL)

The same analyses were replicated in SQL to demonstrate how a data analyst would query a production database for similar insights. The key queries and their interpretations are summarized below.

8.1 Purpose of SQL Analysis

SQL analysis was conducted to:

- Validate Python EDA findings
- Perform complex analytical queries efficiently
- Demonstrate database skills for data analysis
- Create reusable queries for ongoing business intelligence

8.2 Key SQL Queries and Interpretations

8.2.1 Basic Checks and Data Validation

Query 1: Column Information

```
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'retail_full';
```

Purpose: Verify data types after import to ensure compatibility with analysis needs.

Query 2: Record Counts

```
SELECT COUNT(*) FROM retail_full; -- 519,589
SELECT COUNT(*) FROM retail_customers; -- 387,844
```

Interpretation: Confirms cleaning reduced records by 4.1% (22,320 rows) and missing Customer IDs account for 131,745 rows (25.4% of original).

8.2.2 Missing Values Check

```
SELECT
  COUNT(*) FILTER (WHERE Description IS NULL) AS missing_description,
  COUNT(*) FILTER (WHERE Customer_ID IS NULL) AS missing_customer
FROM retail_full;
```

Result: 0 missing values after cleaning.

8.2.3 Negative Values Check

```
SELECT *
FROM retail_full
WHERE Quantity < 0 OR Revenue < 0;
```

Result: No negative values after cleaning phase.

8.2.4 Duplicate Removal

```
DELETE FROM retail_full
WHERE ctid IN (
    SELECT ctid FROM (
        SELECT ctid,
        ROW_NUMBER() OVER (
            PARTITION BY Invoice_No, Stock_Code, Quantity
            ORDER BY Invoice_Date
        ) AS rn
        FROM retail_full
    ) t
    WHERE rn > 1
);
```

Interpretation: Used PostgreSQL's ctid (physical row identifier) to efficiently remove duplicates while preserving one copy of each unique transaction line.

8.2.5 Basic Business Metrics

```
SELECT
    COUNT(DISTINCT Customer_ID) AS total_customers,
    COUNT(DISTINCT Invoice_No) AS total_orders,
    SUM(Revenue) AS total_revenue,
    AVG(Revenue) AS avg_order_value
FROM retail_customers;
```

Results:

- Total Customers: 4,338
- Total Orders: 18,531
- Total Revenue: £8,844,544
- Average Order Value: £477.28

Interpretation: These metrics form the foundation of business performance tracking. The average order value of £477 is relatively high, suggesting significant wholesale/bulk purchases alongside retail sales.

8.2.6 Time Series Analysis in SQL

Monthly Sales Trend:

```
SELECT Month, SUM(Revenue) AS total_revenue
FROM retail_full
GROUP BY Month
ORDER BY Month;
```

Interpretation: Confirms Python findings of steady growth with November peak.

Hourly Sales Pattern:

```
SELECT Hour, SUM(Revenue) AS revenue
FROM retail_full
GROUP BY Hour
ORDER BY Hour;
```

Interpretation: Peak at 10:00 and 12:00 confirms business hours drive most sales.

8.2.7 Country Analysis

```
SELECT Country,
SUM(Quantity) AS quantity,
COUNT(DISTINCT Customer_ID) AS customers,
SUM(Revenue) AS revenue
FROM retail_customers
GROUP BY Country
ORDER BY revenue DESC;
```

Results:

- UK dominates (77% of revenue)
- Netherlands, Ireland, Germany form secondary markets
- Revenue concentration matches Python findings

8.2.8 Product Price Sensitivity Analysis

```
SELECT
Description,
CORR(Unit_Price, Quantity) AS price_sensitivity
FROM retail_full
GROUP BY Description
HAVING COUNT(*) > 50
ORDER BY price_sensitivity;
```

Interpretation:

- Negative correlation for most products (higher price → lower quantity)
- Some products show positive correlation (premium products with high demand)
- Useful for pricing strategy decisions

8.2.9 RFM Analysis in SQL

The RFM analysis SQL query uses:

- Window Functions: NTILE(5) for creating quintiles
- CTEs: For organizing complex logic
- Date Functions: For recency calculation
- CASE Statements: For customer segmentation

RFM Segmentation Logic:

CASE

WHEN r_score = 5 AND f_score = 5 AND m_score = 5 THEN 'Champions'

WHEN r_score >= 4 AND f_score >= 4 THEN 'Loyal Customers'

WHEN r_score >= 3 AND f_score >= 3 THEN 'Potential Loyalists'

WHEN r_score <= 2 AND f_score >= 4 THEN 'At Risk'

WHEN r_score <= 2 AND f_score <= 2 THEN 'Lost Customers'

ELSE 'Average Customers'

END AS customer_segment

Interpretation of RFM Results:

- Champions: Most valuable customers; high recency, frequency, monetary
- Loyal Customers: Consistent buyers; should be rewarded
- Potential Loyalists: Recent buyers with moderate frequency; nurture them
- At Risk: Former frequent buyers who haven't purchased recently; re-engage
- Lost Customers: Low probability of return; low-cost retention efforts

8.2.10 Cohort Analysis in SQL

Cohort Definition:

```
df_customers['CohortMonth'] = df_customers.groupby('Customer_ID')['Invoice_Date'].transform('min').dt.to_period('M')
```

```
df_customers['CohortIndex'] = (df_customers['InvoiceMonth'] - df_customers['CohortMonth']).apply(lambda x: x.n)
```

SQL Equivalent for Retention Rate:

```
SELECT *,
ROUND(
```

```

    users * 100.0 / FIRST_VALUE(users) OVER (PARTITION BY CohortMonth ORDER BY
CohortIndex),
2) AS retention_rate
FROM (
    SELECT CohortMonth,
    CohortIndex,
    COUNT(DISTINCT Customer_ID) AS users
    FROM retail_customers
    GROUP BY CohortMonth, CohortIndex
) t;

```

Interpretation:

- Month 0: Always 100% (first purchase)
- Month 1: Sharp drop (20-40% retention)
- Month 3-6: Stabilization (15-25% retention)
- Month 12: Long-term retention (5-10%)

Business Meaning:

- The first month is critical for retention
- Long-term retention is relatively low but consistent
- Focus on early engagement to improve retention

8.2.11 Revenue Concentration Analysis

```

SELECT
    SUM(Revenue) FILTER (WHERE rank <= 10) * 100.0 / SUM(Revenue) AS top_10_pct
FROM (
    SELECT Customer_ID, SUM(Revenue) AS Revenue,
           RANK() OVER (ORDER BY SUM(Revenue) DESC) AS rank
    FROM retail_customers
    GROUP BY Customer_ID
) t;

```

Interpretation: Top 10 customers account for approximately 15-20% of total revenue, demonstrating significant revenue concentration and highlighting the importance of key account management.

8.2.12 Order Cancellation Detection

```

SELECT *
FROM retail_full
WHERE Invoice_No LIKE 'C%';

```

Purpose: Identify cancelled orders for quality control and operational analysis.

9. KEY FINDINGS AND INTERPRETATIONS

The combination of Python and SQL analysis yielded several critical insights:

1. **Strong UK Market:** The business is heavily reliant on the UK market, which accounts for the majority of revenue and orders. International presence exists but is secondary.
2. **Peak Sales Times:** Sales are highest on **Thursdays** and during the **10 AM to 3 PM hour**. This suggests optimal times for marketing, promotions, and resource allocation.
3. **Drivers of Revenue:** Revenue is almost perfectly correlated with quantity, not price. Increasing average basket size is more impactful than raising prices.
4. **Product Mix:** The top-selling products are low-priced novelty items, while the top-revenue products are higher-priced items. A balanced inventory strategy is needed to cater to both volume and value drivers.
5. **High-Value Customers:** A small percentage of customers generate a large portion of revenue. Identifying and retaining these customers is crucial.
6. **Loyalty and Retention:**
 - The repeat customer rate is a healthy **65%**.
 - RFM analysis provides a clear framework to target different customer segments (Champions, At Risk, etc.).
 - Cohort analysis reveals that the **December 2010 cohort** is significantly more loyal than later cohorts, warranting an investigation into what made them so.
7. **Data Quality:** The significant number of missing CustomerID values (25%) is a major data quality issue that limits the depth of customer-level analysis.

9.1 Business Performance Summary

Metric	Value
Total Revenue (Full Dataset)	£10,587,082
Total Revenue (Valid Customers)	£8,844,544
Total Orders	19,959
Unique Customers	4,338
Average Order Value	£530.44
Total Products Sold	3,922
Countries Served	38

9.2 Customer Insights

1. **Customer Concentration:**
 - Top 10 customers generate 15-20% of total revenue

- High-value customers (top 20%) contribute 80% of revenue (Pareto principle)
- Customer 14646 spent over £280,000 – extremely high-value
- 2. **Customer Loyalty:**
 - 65.6% of customers are repeat buyers
 - 34.4% are one-time purchasers
 - Strong retention but opportunity to improve
- 3. **Customer Segmentation (RFM):**
 - **Champions (555):** Best customers – reward and retain
 - **Loyal Customers:** Engaged buyers – cross-sell opportunities
 - **At Risk:** Declining engagement – re-engagement campaigns needed
 - **Lost:** Low priority for acquisition, focus on cost-effective retention

9.3 Sales Patterns

1. **Temporal Patterns:**
 - **Monthly:** Steady growth, November peak (holiday shopping)
 - **Weekly:** Mid-week peak (Tuesday-Thursday), weekend slowdown
 - **Hourly:** Business hours (9:00-17:00), lunch hour secondary peak
2. **Product Performance:**
 - Volume leader: "PAPER CRAFT, LITTLE BIRDIE" (80,995 units)
 - Gift items and party supplies are top sellers
 - Retro-spot design products are consistently popular
3. **Geographical Distribution:**
 - United Kingdom dominates (77.4% of revenue)
 - Netherlands, Ireland, Germany are top international markets
 - 38 countries total – potential for international expansion

9.4 Product Mix Analysis

Category	Characteristics	Strategy
High Volume, Low Price	T-light holders, napkins	Maintain availability, volume discounts
High Value, Low Volume	Cake stands, gift sets	Premium positioning, bundled offers
Wholesale Items	Paper crafts, bulk items	B2B marketing, bulk discounts

9.5 Retention Insights (Cohort Analysis)

- **First month drop-off:** 60-80% of customers don't return
- **Stabilization:** After 3 months, retention stabilizes at 15-25%
- **Long-term retention:** 5-10% at 12 months
- **Best cohorts:** December 2010 showed highest long-term retention (27% at 12 months)

Implication: Critical to engage customers in first 30 days post-purchase to improve retention.

9.6 RFM Segmentation Recommendations

Segment	% of Customers	Strategy
Champions	5%	VIP treatment, exclusive previews, loyalty rewards
Loyal	10%	Cross-sell, loyalty program, referral incentives
Potential Loyalists	15%	Welcome series, educational content, small incentives
At Risk	20%	Re-engagement emails, special offers, feedback surveys
Lost	35%	Limited re-engagement, focus on cost efficiency

9.7 Business Recommendations

Short-term Actions (0-3 months)

- 1. Implement Customer Retention Program**
 - Welcome email sequence for new customers
 - First purchase follow-up after 30 days
 - Birthday/anniversary offers
- 2. Optimize Marketing Timing**
 - Schedule campaigns for Tuesday-Thursday (peak days)
 - Focus email sends for 10:00 and 12:00 (peak hours)
 - Weekend promotions to boost slower days
- 3. Address Data Quality Issues**
 - Fix missing Customer ID (24.9% of records)
 - Implement better tracking for guest checkouts
 - Regular data quality audits

Medium-term Actions (3-6 months)

- 1. Develop Loyalty Program**

- Tiered rewards based on RFM scores
 - Points for purchases, referrals, reviews
 - Exclusive benefits for top segments
2. **International Market Expansion**
 - Focus on Netherlands, Ireland, Germany
 - Localized marketing campaigns
 - Consider local fulfillment centers
 3. **Product Portfolio Optimization**
 - Promote high-margin gift sets
 - Bundle popular items
 - Discontinue underperforming products

Long-term Actions (6-12 months)

1. **Predictive Analytics Implementation**
 - Customer churn prediction model
 - Next purchase prediction
 - Personalized recommendations
2. **Automated Marketing Campaigns**
 - RFM-triggered email campaigns
 - Abandoned cart recovery
 - Cross-sell recommendations based on purchase history
3. **Customer Lifetime Value Optimization**
 - Allocate marketing budget based on predicted CLV
 - VIP account management for top customers
 - Acquisition cost limits based on CLV

9.8 Python vs SQL Analysis Comparison

Aspect	Python (Google Colab)	SQL (PostgreSQL)
Data Volume	Handles 500K+ rows efficiently	Excellent for large datasets
Visualization	Rich visualization options	Limited, requires external tools
Complex Logic	Easy to implement custom logic	Powerful with window functions
Performance	Good for moderate datasets	Optimized for large databases
Reusability	Code must be rerun	Queries can be saved as views
Learning Curve	Moderate	Moderate to advanced

Findings: Both approaches validated each other's results, confirming data quality and analytical accuracy.

10. LIMITATIONS AND CHALLENGES

10.1 Data Quality Limitations

1. Missing Customer IDs (24.9%)

- **Impact:** Unable to track approximately 1/4 of transactions to specific customers
- **Consequence:** Incomplete customer behavior analysis
- **Mitigation:** Created separate full and customer datasets for different analyses

2. Missing Product Descriptions (0.27%)

- **Impact:** Removed these records (1,454 rows) from analysis
- **Consequence:** Minor loss of transaction data
- **Mitigation:** Acceptable loss given small percentage

3. Negative Quantities and Unit Prices

- **Impact:** Required filtering of records with negative values
- **Consequence:** Some cancelled/returned transactions excluded
- **Mitigation:** Documented handling approach

10.2 Analytical Limitations

1. Log Transformation Interpretation

- **Challenge:** Log-transformed data requires careful interpretation
- **Solution:** Presented both normal and log-scale visualizations
- **Benefit:** Better understanding of underlying distributions

2. Outlier Impact

- **Challenge:** Extreme values (e.g., 80,995 units) skew averages
- **Solution:** Used median and percentile-based analysis alongside means
- **Benefit:** More robust insights

3. Cohort Analysis Data Requirements

- **Challenge:** Requires complete purchase history for accurate cohorts
- **Limitation:** Dataset only covers 12 months; some cohorts truncated
- **Mitigation:** Acknowledged in interpretation

10.3 Technical Challenges

1. Data Volume Handling

- **Challenge:** 541,909 rows initially loaded
- **Solution:** Used pandas efficiently, indexed columns for SQL queries

- **Outcome:** Smooth processing without memory issues
- 2. **Data Type Inconsistencies**
 - **Challenge:** Customer ID as float64 with decimal points
 - **Solution:** Converted to Int64 then string
 - **Outcome:** Consistent data types across datasets
- 3. **Date Format Variations**
 - **Challenge:** Inconsistent date formatting in raw data
 - **Solution:** Standardized using pandas datetime functionality
 - **Outcome:** Reliable temporal analysis

10.4 Business Context Limitations

1. **Single Business Type**
 - **Limitation:** Findings specific to online retail gift company
 - **Generalizability:** Limited to similar business models
 - **Context:** Results should be interpreted within this context
2. **Limited Time Period**
 - **Limitation:** Only 12 months of data
 - **Impact:** Cannot analyze full yearly seasonality patterns
 - **Mitigation:** Focused on available patterns within period
3. **No External Factors**
 - **Limitation:** Cannot account for marketing campaigns, economic conditions
 - **Impact:** Trends may be influenced by unmeasured factors
 - **Acknowledgment:** Noted in interpretations

10.5 Lessons Learned

1. **Data Quality is Critical:** 24.9% missing Customer IDs significantly impacts analysis depth
2. **Multiple Data Views:** Creating separate datasets for different analytical purposes is essential
3. **Transformation Matters:** Log transformations reveal patterns hidden in raw data
4. **Validation is Key:** SQL validation confirmed Python findings, increasing confidence
5. **Business Context:** Technical findings must be translated to actionable business insights

11. CONCLUSION

This project successfully performed a comprehensive Exploratory Data Analysis on the Online Retail dataset. By combining the strengths of Python for advanced visualization and statistical analysis with the power of SQL for structured querying and operational insights, we were able to uncover a wealth of actionable business intelligence.

The analysis revealed that the UK-based online retailer has a strong customer base with high repeat rates but is heavily dependent on a small number of high-value customers and the UK market. Clear patterns in sales timing provide direct levers for operational optimization. The application of RFM and Cohort analyses has provided a robust framework for customer segmentation and retention tracking, which are essential for long-term business growth.

The findings from this EDA can directly inform marketing strategies (targeting specific customer segments), operational planning (staffing for peak hours), and inventory management (balancing high-volume and high-revenue products). The limitations, particularly the missing Customer IDs, highlight areas for data quality improvement. Overall, this project demonstrates the immense value of data-driven decision-making and the power of EDA as a foundational step in any data science workflow.

11.1 Summary of Achievements

This project successfully conducted a comprehensive Exploratory Data Analysis on the Online Retail Dataset, achieving all stated objectives:

1. **Data Understanding and Preparation:** Completed thorough data cleaning, handled missing values, removed duplicates, and corrected data types for 541,909 records.
2. **Customer Behavior Analysis:** Identified customer segments through RFM analysis, revealed that 65.6% of customers are repeat buyers, and quantified revenue concentration among top customers.
3. **Sales Performance Analysis:** Uncovered temporal patterns including November peak sales, mid-week peaks, and business-hour sales concentration.
4. **Geographical Analysis:** Confirmed UK dominance (77.4% of revenue) and identified top international markets.
5. **Advanced Analytical Modeling:** Successfully implemented RFM analysis, Cohort analysis, and CLV estimation, providing deep insights into customer value and retention.
6. **Actionable Business Insights:** Generated 12+ actionable recommendations across short, medium, and long-term timeframes.

11.2 Key Takeaways

Area	Key Finding	Business Implication
Revenue	£10.6M total, £530 AOV	Strong business performance
Customers	65.6% repeat rate	Good loyalty, room for improvement
Concentration	Top 10 customers = 15-20% revenue	Key account management critical
Timing	Mid-week, business hours peak	Optimize marketing timing
Geography	UK dominates (77%)	Expand international
Retention	60-80% drop after month 1	Focus on first 30 days
CLV	£2,040 average	Target high-CLV segments

11.3 Final Recommendations

Immediate Priorities:

1. Fix data quality issues (missing Customer IDs)
2. Implement first 30-day retention program
3. Schedule marketing for peak days/hours
4. Develop key account program for top 10 customers

Strategic Initiatives:

1. Build loyalty program based on RFM segmentation
2. Expand into top international markets
3. Optimize product mix for volume/value balance
4. Implement predictive analytics for personalization

Long-term Vision:

1. Develop CLV-based marketing allocation
2. Build automated, personalized marketing campaigns
3. Create data infrastructure for real-time analytics
4. Establish customer-centric KPIs and monitoring

11.4 Project Significance

This project demonstrates the complete data science workflow from raw data to actionable business insights. It showcases:

- **Technical Skills:** Python, pandas, SQL, data visualization, statistical analysis
- **Analytical Skills:** RFM analysis, cohort analysis, CLV calculation, segmentation

- **Business Acumen:** Translating technical findings to strategic recommendations
- **Communication:** Clear interpretation and visualization of complex data

The findings provide a solid foundation for data-driven decision-making in the online retail business and demonstrate the value of thorough exploratory data analysis in understanding customer behavior and business performance.

12. REFERENCES

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- **GOOGLE COLAB NOTEBOOK LINK OF THE EDA ON THE ONLINE RETAIL DATASET:**

<https://colab.research.google.com/drive/1stOR1i76hwvuVJ7R2JU3iSoWlebMOtu?usp=sharing>

13. APPENDIX

- **Appendix:** Full Python code and outputs from the Google Colab notebook, SQL code used in pgAdmin4, Screenshots of SQL query outputs from pgAdmin4.